

# Chapter 1: Analytical Foundations

## Introduction & Mappings and Functions

Advanced Mathematics  
(Tongji University, 8th Ed.)

## 1 Introduction to the Course

## 2 Mappings and Functions

- Functions
- Properties of Functions
- Inverse and Composite Functions
- Elementary Functions
- Worked Examples
- Summary & Challenge Problems

## Main Textbook

*Advanced Mathematics* (8th Edition)

Department of Applied Mathematics, Tongji University

Higher Education Press, 2020.

# Textbook & References

## Main Textbook

*Advanced Mathematics* (8th Edition)

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Higher Education Press, 2020.

## Supplementary Reading

*Solutions Manual for Advanced Mathematics (8th Ed., Combined Volume)*

China Water & Power Press, 2021.

## ① **Analytical Foundations:** Functions, Limits, Continuity

# Course Content Overview

- ① **Analytical Foundations:** Functions, Limits, Continuity
- ② **Calculus of One Variable** (Volumes I & II)

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Multivariable Calculus is covered in Volumes III–V.

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## Marx & Engels on Mathematics

“To understand nature dialectically and materially, one must be familiar with mathematics.”

— Engels

“A science reaches true perfection only when it employs mathematics.”

— Marx

# Section 1: Mappings and Functions

I. Mappings      II. Functions

# Motivating Examples for Mappings

## Example (Addition of $x$ and $\sin x$ )

For every  $x \in \mathbb{R}$ ,

$$y = x + \sin x$$

defines a mapping from  $\mathbb{R}$  to  $\mathbb{R}$ .

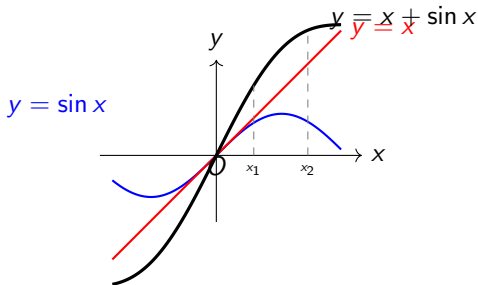
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## Example: Projection onto the $y$ -axis

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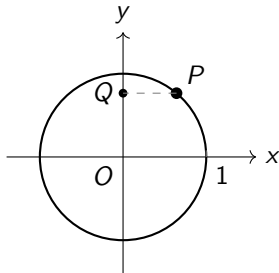
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For every point  $P \in C$ , project orthogonally onto the  $y$ -axis to obtain  $Q \in Y$ . This defines a mapping  $C \rightarrow Y$ .



### Properties:

- This mapping is **surjective**: every  $Q \in Y$  has a pre-image.
- It is **not injective**: two symmetric points  $P, P'$  map to the same  $Q$ .

# Definition of a Mapping

## Definition (Mapping)

Let  $X$  and  $Y$  be two nonempty sets. If there exists a rule  $f$  such that

$$\forall x \in X, \exists \text{ a unique } y \in Y \text{ corresponding to } x,$$

then  $f$  is called a **mapping** from  $X$  to  $Y$ , denoted

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- $y = f(x)$ : the **image** of  $x$  under  $f$ .
- $x$ : a **pre-image** of  $y$  under  $f$  (not necessarily unique!).
- $X$ : the **domain** of  $f$ .
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## Three Elements of a Mapping

Domain, Correspondence Rule, Range.

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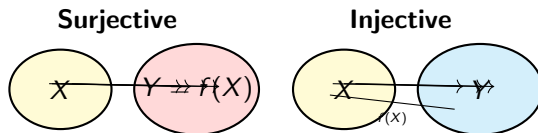
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# Examples of Mappings (I)

## Example (Heron's Formula — Surjective)

For every triangle  $\Delta$  in the set of all triangles,

$$S = \sqrt{p(p-a)(p-b)(p-c)}, \quad p = \frac{1}{2}(a+b+c)$$

maps  $\Delta$  to its area  $S \in (0, +\infty)$ . This mapping is **surjective**.

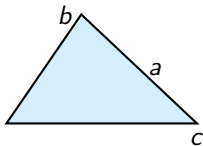
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## Example (Area Under $e^x$ — Surjective)

For each  $x \in [0, +\infty)$ :

$$S(x) = \int_0^x e^t dt = e^x - 1.$$

$S \in [0, +\infty)$  is surjective onto itself.

# Examples of Mappings (II)

## Example (Polar-to-Cartesian — Surjective)

$$f: [0, +\infty) \times [0, 2\pi) \longrightarrow \mathbb{R}^2, \quad \begin{cases} x = r \cos \theta, \\ y = r \sin \theta, \end{cases}$$

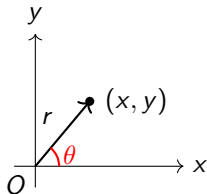
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## Key Point

A **function** is precisely a mapping whose domain is a subset of  $\mathbb{R}$  and whose codomain is  $\mathbb{R}$ .

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## Important

Only **injective** mappings admit an inverse!

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- Domain:  $D_{f^{-1}} = [-1, 1]$
- Range:  $R_{f^{-1}} = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

# Composite Mapping

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Given two mappings  $g: X \rightarrow Y_1$  and  $f: Y_2 \rightarrow Z$  with  $Y_1 \subseteq Y_2$ , we define a mapping from  $X$  to  $Z$  by sending each  $x \in X$  to  $f[g(x)] \in Z$ .

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## Condition for Composition

The range of  $g$  must be contained in the domain of  $f$ :

$$R_g \subseteq D_f.$$

# Example: Composition Order Matters

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## Note

Composition is **not commutative**: order matters!

# Definition of a Function

## Definition (Function)

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## Graph of a Function

$$C = \{(x, y) \mid y = f(x), x \in D\} \subset D \times f(D)$$

# Ways to Represent a Function

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## Example (Absolute Value Function)

$$y = |x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases} \quad \text{Domain } (-\infty, +\infty), \text{ Range } [0, +\infty)$$

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**Domain convention:** For functions without a practical context, the domain is taken as the largest set where the expression makes sense. For applied problems, the domain must be stated explicitly.

## Example 3: Piecewise Function

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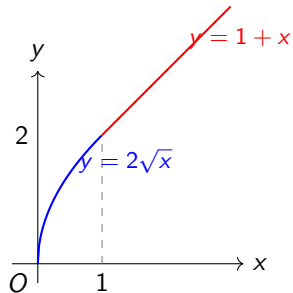
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$$f(\frac{1}{t}) = \begin{cases} 1 + \frac{1}{t}, & 0 < t < 1, \\ \frac{2}{\sqrt{t}}, & t \geq 1. \end{cases}$$

## Example 3: Graph of the Piecewise Function



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## Related Concepts

One can also define *bounded above*, *bounded below*, etc. (see p. 7).

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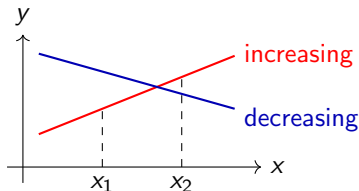
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## Definition (Parity)

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For all  $x \in D$  such that  $-x \in D$ :

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## Example (Hyperbolic Cosine — Even)

$$y = \operatorname{ch} x = \frac{e^x + e^{-x}}{2}, \quad \operatorname{ch}(-x) = \operatorname{ch} x.$$

# Hyperbolic Functions (cont.)

## Example (Hyperbolic Sine — Odd)

$$y = \operatorname{sh} x = \frac{e^x - e^{-x}}{2}, \quad \operatorname{sh}(-x) = -\operatorname{sh} x.$$

# Hyperbolic Functions (cont.)

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## Even–Odd Decomposition

Any function  $f$  defined on  $(-l, l)$  can be decomposed as

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even part}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd part}}.$$

## Definition (Periodic Function)

If  $\forall x \in D, \exists l > 0$  with  $x \pm l \in D$  such that

$$f(x \pm l) = f(x),$$

then  $f$  is called a **periodic function** and  $l$  is a **period**  
(usually meaning the *smallest positive period*).

# Periodicity

## Definition (Periodic Function)

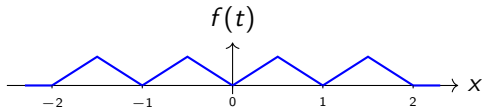
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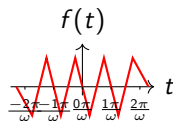
$$y = |\sin x|$$

period =  $\pi$



$$y = \sin(\omega t)$$

period =  $\frac{2\pi}{\omega}$



# Periodicity: Cautionary Notes

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Not every periodic function has a smallest positive period!

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## Example (Constant Function)

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## Example (Dirichlet Function)

$$f(x) = \begin{cases} 1, & x \text{ is rational,} \\ 0, & x \text{ is irrational,} \end{cases}$$

is periodic with *every rational* as a period — no smallest positive period exists.

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## Example

$y = e^x$  (exponential) and  $y = \ln x$  (logarithm) are inverses of each other. Both are strictly increasing; their graphs are symmetric about  $y = x$ .

# Composite Functions

Given a function chain

$$x \text{ ① } u \text{ ② } y,$$

the composition yields  $y = f[g(x)]$ , where  $u$  is the **intermediate variable**.

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## Example

$y = \sqrt{u}$ ,  $u = x^2 - 1$  cannot compose over  $\mathbb{R}$  (domain issue).

But restricting  $u \geq 0$  (i.e.  $|x| \geq 1$ ) gives a valid composite  $y = \sqrt{x^2 - 1}$ .

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## Convention

When writing a composite function, the domain is often omitted by default — it is understood that each link in the function chain satisfies the composition condition.

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- ⑤ **Inverse trigonometric functions:**  $\arcsin x, \arccos x, \arctan x$ , etc.

# Elementary & Non-elementary Functions

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A function obtained from constants and basic elementary functions by a **finite** number of arithmetic operations and compositions, expressible as a single formula, is called an **elementary function**.

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## Non-elementary Functions

Functions that cannot be written as a single formula (piecewise-defined, etc.)

# Examples of Non-elementary Functions

## Example (Signum Function)

$$y = \operatorname{sgn} x = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases}$$

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$$y = [x] = n, \quad \text{when } n \leq x < n + 1, \quad n \in \mathbb{Z}.$$

These are defined piecewise and have no single analytic expression valid everywhere.

## Example 4: Composite of a Piecewise Function

### Example

Let  $f(x) = \begin{cases} 3x + 1, & x < 1, \\ x, & x \geq 1. \end{cases}$  Find  $f[f(x)]$ .

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Substitute  $x \mapsto f(x)$  into the definition of  $f$ :

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Condition  $f(x) < 1$  means  $3x + 1 < 1$ , i.e.  $x < 0$ . When  $x < 0$ :  $f[f(x)] = 3(3x + 1) + 1 = 9x + 4$ .

When  $0 \leq x < 1$ :  $f[f(x)] = 3x + 1$ .

When  $x \geq 1$ :  $f[f(x)] = x$ .

## Example 4: Final Answer

Result

$$f[f(x)] = \begin{cases} 9x + 4, & x < 0, \\ 3x + 1, & 0 \leq x < 1, \\ x, & x \geq 1. \end{cases}$$

## Example 4: Final Answer

### Result

$$f[f(x)] = \begin{cases} 9x + 4, & x < 0, \\ 3x + 1, & 0 \leq x < 1, \\ x, & x \geq 1. \end{cases}$$

### Key Technique

For piecewise functions, handle each branch separately and carefully track the conditions after substitution.

## Example 5: Inverse of a Piecewise Function

### Example

Find the inverse of  $y = \begin{cases} x^2, & -1 \leq x < 0, \\ \ln x, & 0 < x \leq 1, \\ 2e^{x-1}, & 1 < x \leq 2. \end{cases}$

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**Branch 3:**  $1 < x \leq 2, y = 2e^{x-1} \in (2, 2e] \Rightarrow x = 1 + \ln \frac{y}{2}, y \in (2, 2e]$ .

## Example 5: Inverse Function

### Inverse Function

$$y = f^{-1}(x) = \begin{cases} e^x, & x \in (-\infty, 0], \\ -\sqrt{x}, & x \in (0, 1], \\ 1 + \ln \frac{x}{2}, & x \in (2, 2e]. \end{cases}$$

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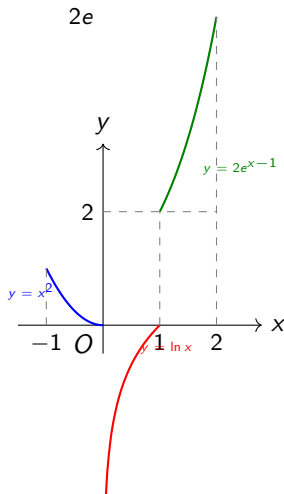
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### Note

The domain of  $f^{-1}$  equals the range of  $f$ . Each branch of the inverse corresponds to one monotone branch of  $f$ .

## Example 5: Graph of the Original Function



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- ④ **Inverse & composite functions:** existence conditions, properties, techniques for piecewise cases.

# Challenge Problem 1: Proving Oddness

## Example

Suppose  $f(0) = 0$  and for  $x \neq 0$ :

$$a f(x) + b f\left(\frac{1}{x}\right) = \frac{c}{x},$$

where  $a, b, c$  are constants with  $|a| \neq |b|$ . Prove that  $f(x)$  is an odd function.

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## Step 1: Set Up the System

Substitute  $t = \frac{1}{x}$  to obtain a second equation:

$$\begin{cases} a f(x) + b f(1/x) = c/x, \\ a f(1/x) + b f(x) = cx. \end{cases}$$

# Challenge Problem 1: Solution (cont.)

## Step 2: Eliminate $f(1/x)$

Solving the linear system for  $f(x)$ :

$$f(x) = \frac{c}{b^2 - a^2} \left( bx - \frac{a}{x} \right) \quad (x \neq 0).$$

# Challenge Problem 1: Solution (cont.)

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## Conclusion

Clearly  $f(-x) = -f(x)$  and  $f(0) = 0$ .

Therefore  $f(x)$  is an **odd function**. ■

## Challenge Problem 2: Symmetry Implies Periodicity

### Example

Let  $y = f(x)$ ,  $x \in (-\infty, +\infty)$ . If the graph is symmetric about both  $x = a$  and  $x = b$  ( $a \neq b$ ), prove that  $f(x)$  is periodic.

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## Step 1: Symmetry Conditions

$$f(a + x) = f(a - x), \quad f(b + x) = f(b - x).$$

## Step 2: Apply Symmetry Twice

$$\begin{aligned} f(x) &= f[a + (x - a)] = f[a - (x - a)] = f(2a - x) \\ &= f[b + (2a - x - b)]. \end{aligned}$$

## Challenge Problem 2: Proof (cont.)

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### Conclusion

Hence  $f(x)$  is periodic with **period**  $T = 2(b - a)$ . ■

*Key idea:* Two reflections about distinct vertical axes compose to a horizontal translation.

# End of Section 1

Mappings and Functions